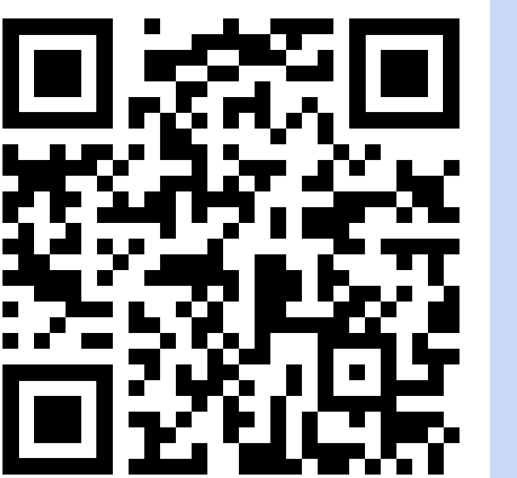


Are Latent Reasoning Models Easily Interpretable?

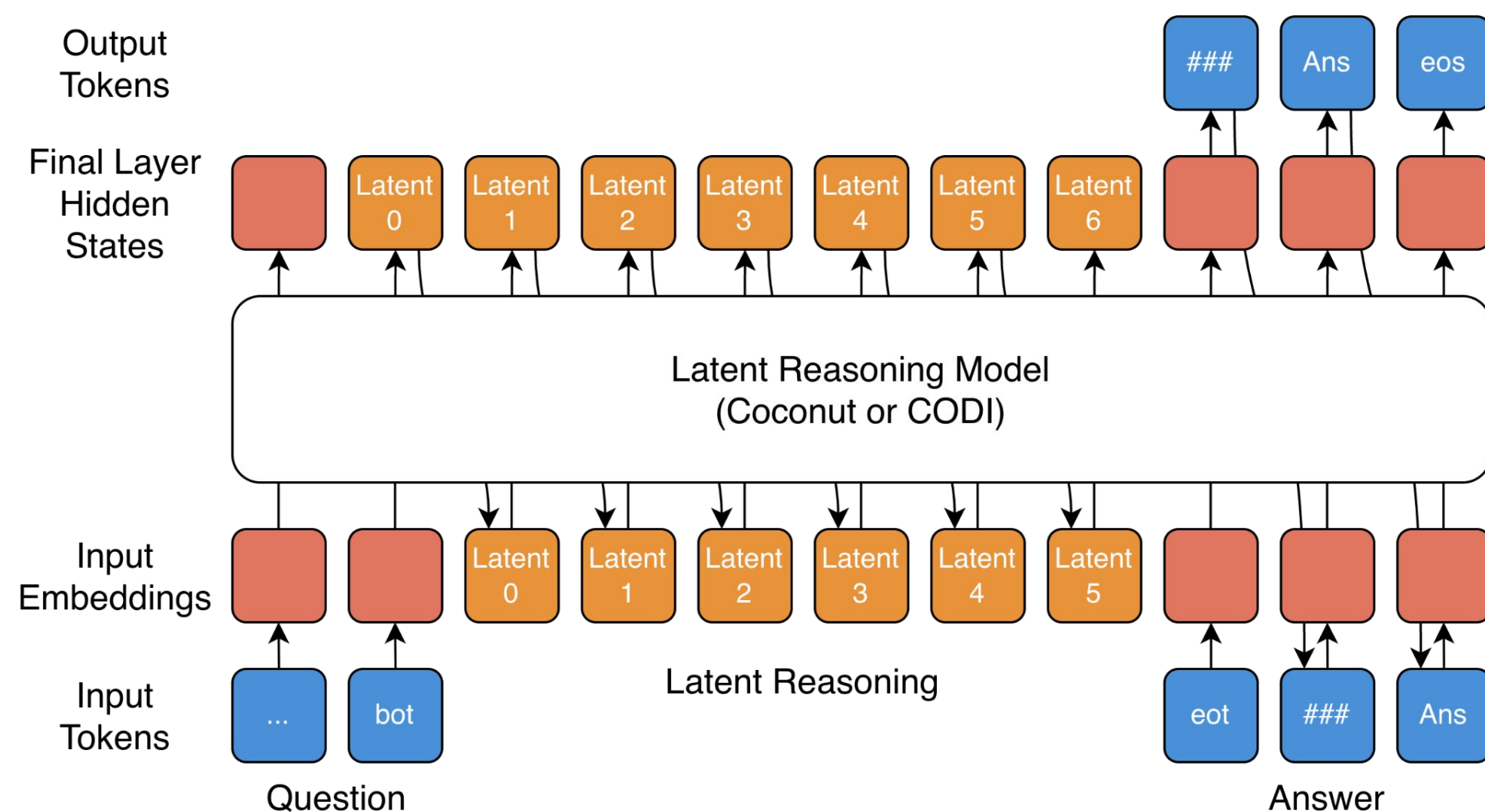


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Takeaways

- Latent reasoning models (width-based) don't fully use their reasoning tokens. On graph reachability datasets, they don't use any.
- When latent tokens are used, latent reasoning models often encode expected reasoning traces
- We can often extract these NL reasoning traces without supervision



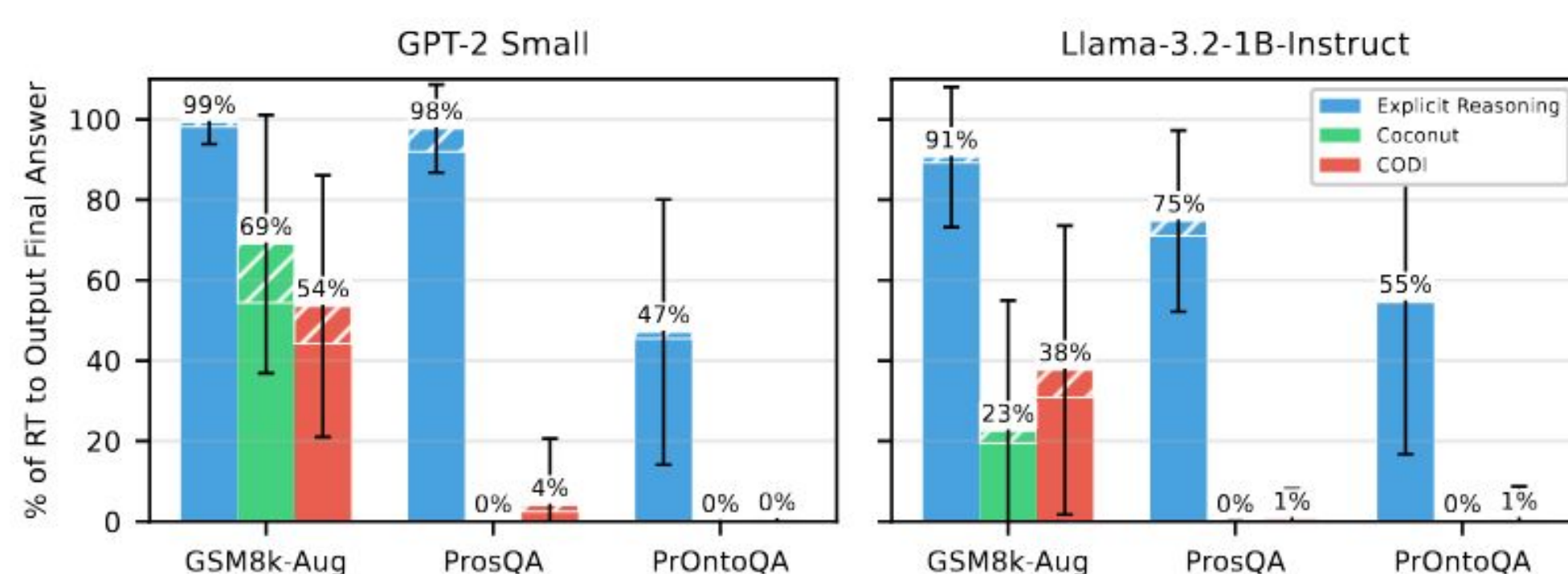
Motivation

- Latent reasoning models reason in a continuous vector space, making them difficult to monitor and audit
- Goal: Investigate whether current LRMs can be interpreted using simple interpretability techniques, and whether their latent tokens are even necessary for performance

Latent reasoning tokens are often unnecessary for Model Performance

- We use early stopping to determine how many reasoning tokens are needed
- Previous work^{1,2} has argued LRMs search through multiple solutions paths in parallel

Prompt	bot	0 x latent	eot	85	
Prompt	bot	1 x latent	eot	85	
Prompt	bot	2 x latent	eot	55	First match
Prompt	bot	3 x latent	eot	165	
Prompt	bot	4 x latent	eot	55	Stable match
Prompt	bot	5 x latent	eot	55	
Prompt	bot	6 x latent	eot	55	Final Answer



Q: Why do latent reasoning models show performance gains on graph search datasets?

Method	Base Model	GSM8k-Aug		PrOntoQA		ProsQA	
		Acc. (%)	# Tok.	Acc. (%)	# Tok.	Acc. (%)	# Tok.
No-CoT	GPT-2 Small	16.8 (16.5 [†])	3.2	87.9 (93.8 [†])	3.0	76.0 (76.7 [†])	9.5
CoT	GPT-2 Small	41.6 (42.9 [†])	31.0	99.3 (98.8 [†])	92.7	74.2 (77.5 [†])	51.6
Coconut	GPT-2 Small	33.1 (34.1 [†])	9.2	99.0 (99.8 [†])	9.0	98.0 (97.0 [†])	15.5
CODI	GPT-2 Small	42.2* (43.7 [†])	12.3	95.1	12.0	81.6	18.2
No-CoT	Llama-3.2-1B	30.1 (30.9 [†])	4.2	99.8	3.0	87.8	8.6
CoT	Llama-3.2-1B	59.4 (61.6 [†])	29.7	99.6	85.6	95.2	42.6
Coconut	Llama-3.2-1B	35.7 (45.3 [†])	10.2	98.8	9.0	97.6	14.7
CODI	Llama-3.2-1B	56.0 (55.6 [†])	13.2	93.6	12.0	99.0	17.7

A: When controlling for training data, they perform as well as non-reasoning models

LRM	Base Model	Reasoning Mode	GSM8k-Aug	PrOntoQA	ProsQA
Coconut	GPT-2 Small	No-CoT	0.3	0.0	0.0
		CoT	19.2	0.0	-6.0
	Llama-3.2-1B-Instruct	No-CoT	-0.9	0.0	0.0
		CoT	29.3	-0.3	-1.4
CODI	GPT-2 Small	No-CoT	-5.9	-0.3	-0.2
		CoT	3.5	-0.4	-7.4
	Llama-3.2-1B-Instruct	No-CoT	2.4	0.0	0.4
		CoT	19.9	6.2	-1.6

Relative performance of No-CoT and CoT to latent reasoning, when controlling for training data. Positive values indicate latent reasoning underperforms.

Latent reasoning models often encode expected reasoning traces

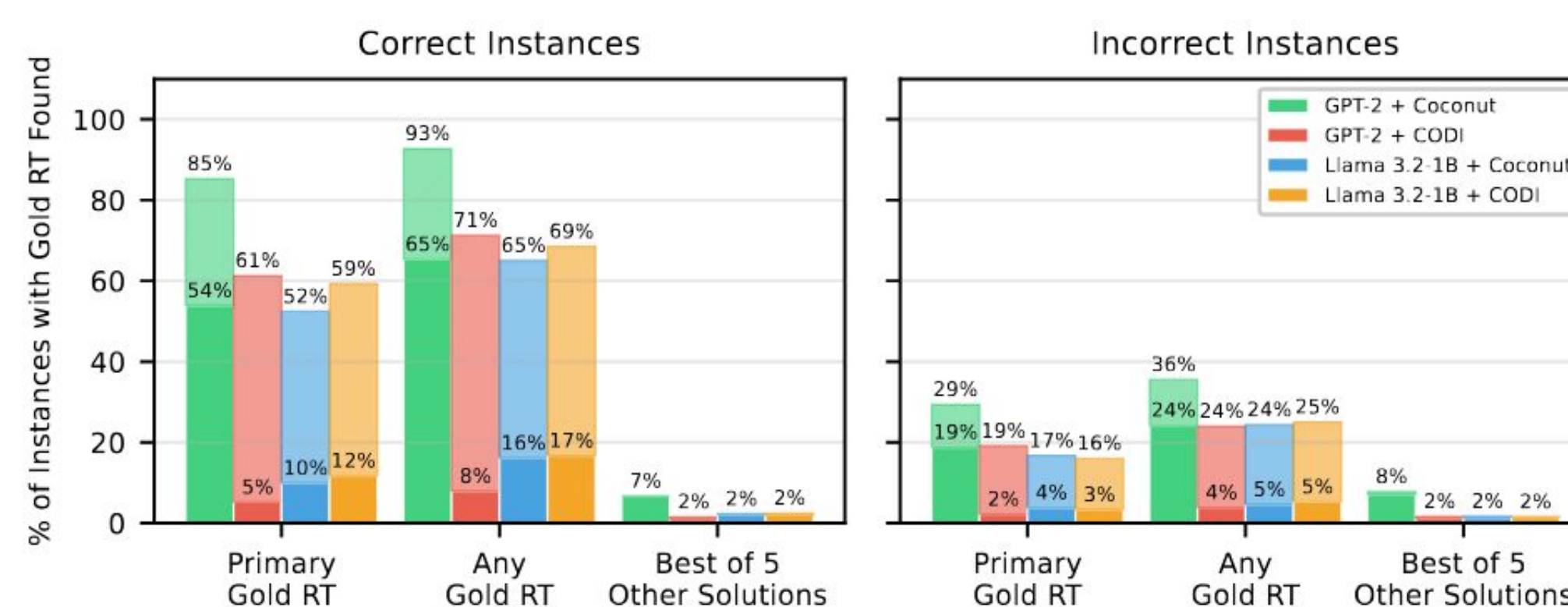
- We can find the expected reasoning trace in the vocabulary projections of latent tokens
- This suggests that latent reasoning models reason in similar ways to explicit reasoning models, instead of arriving at their answer in a completely uninterpretable way

Gold Reasoning Trace:

- $10 \times 3 = 30$
- $5 \times 8 = 40$
- $30 + 40 = 70$

Output	L0	L1	L2	L3	L4	L5	L6	###	70
1	###	30	eos	###	40	###	70	###	70
2	3	20	whereas	8	44	eos	74	90	110
3	10	20	whereas	8	44	eos	74	90	110
10	30	300	plus	20	4	332	370	Two	80
Input	bot	L0	L1	L2	L3	L4	L5	eot	###

Top-10 tokens from logit lens



Can we extract reasoning traces in latent tokens without supervision?

- LRMs have some consistency in their latent tokens structure
- By running multiple variants of the same problem with small perturbations, we find encoded steps in latent tokens

Question: Ali is a superstar counter. He has won 22 medals for counting super fast. His friend Izzy is also a really good counter and has 5 less medals than Ali. Together they have 10 times less medals than have been given out for counting. How many medals have been given out for counting?

Gold Reasoning Trace: «22 - 5 = 17», «22 + 17 = 39», «10 × 39 = 390»

Numbers from the question:

Output	Latent 0	Latent 1	Latent 2	Latent 3	Latent 4	Latent 5	Latent 6	The	answer	is	:	:	:	390
1	(	39	17	.	39	17	atching	The	answer	is	:	:	:	390
2	<<	40	azzo	.	399	18	dra	A	answers	are	:	:	:	380
3	[	37	?	anya	Bre	7	s	<<	answer	.	:	:	:	378
10	combining	ly	ieri	→	393	157	atched	What	response	adalah	:	:	:	39
Input	bot	Latent 0	Latent 1	Latent 2	Latent 3	Latent 4	Latent 5	eot	The	answer	is	:	:	:

Verification Attempt 1 for Candidate 2: «22 - 5 = 17»

Modification 1: Replace 22 with 23

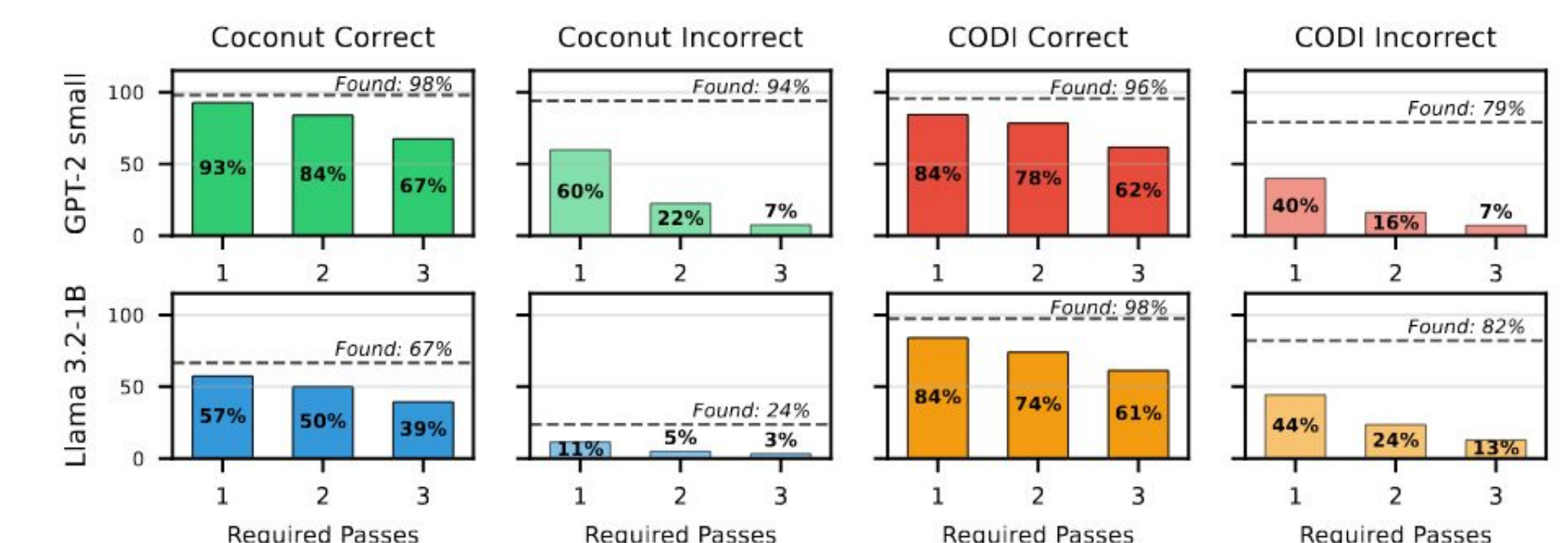
Modified Prompt: Ali is a superstar counter. He has won 23 medals for counting super fast. His friend Izzy is also a really good counter and has 5 less medals than Ali. Together they have 10 times less medals than have been given out for counting. How many medals have been given out for counting?

Candidate 2 with modification 1: «23 - 5 = 18»

Top-1 integer token with modification: 18

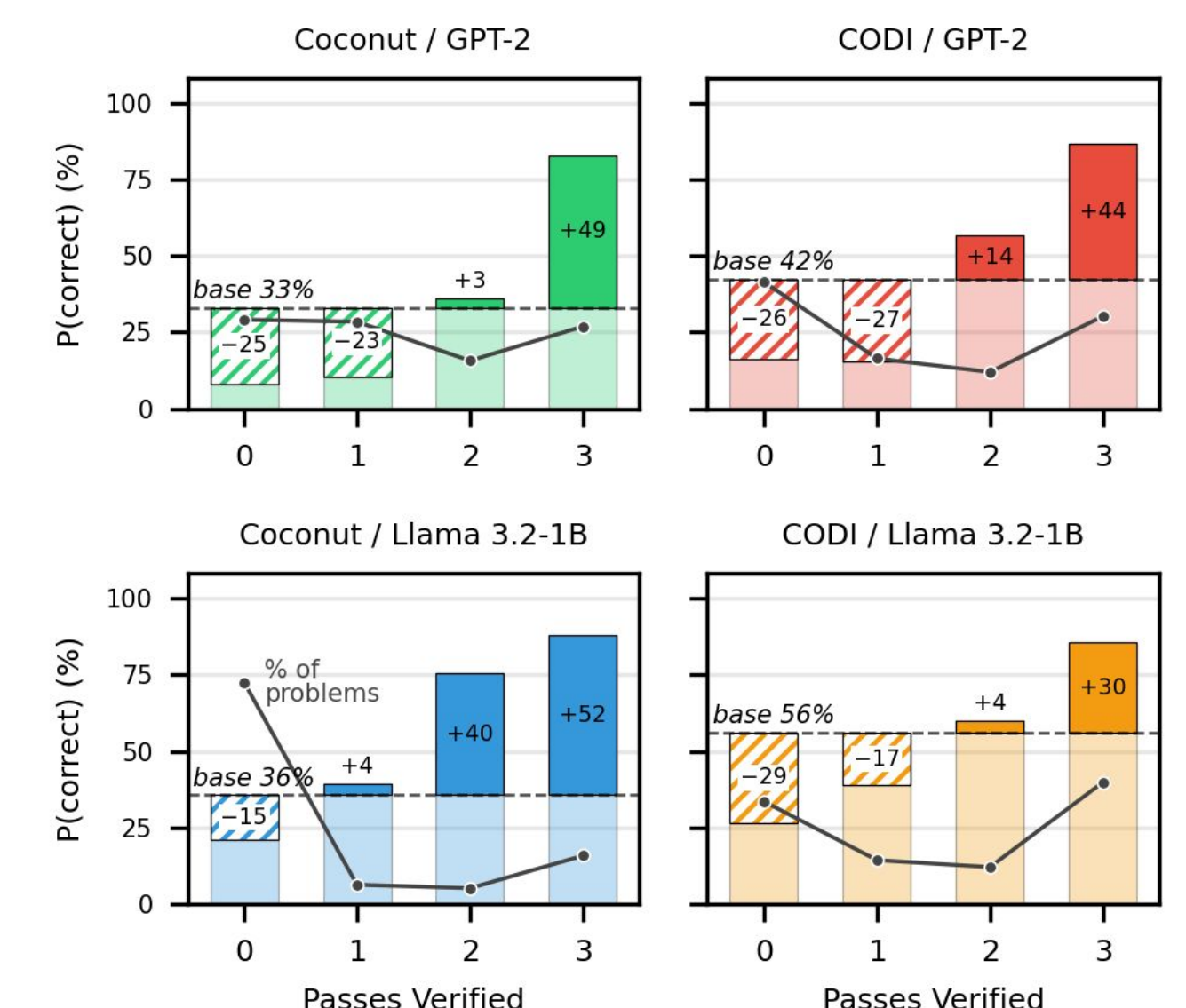
Passes Verification Attempt 1

Output	Latent 0	Latent 1	Latent 2	Latent 3	Latent 4	Latent 5	Latent 6	The	answer	is	:	:	:	410
1	(	41	18	.	41	18	s	The	answer	is	:	:	:	410
2	<<	45	23	eono	PI	17	m	<<	answers	are	:	:	:	420
3	Pair	40	17	.	39	19	sar	<<	answer	.	:	:	:	415
10	"q"	51	20	?	43	38	ived	What	reply	adalah	:	:	:	510
Input	bot	Latent 0	Latent 1	Latent 2	Latent 3	Latent 4	Latent 5	eot	The	answer	is	:	:	:



Interpretability as a signal of prediction correctness

- Extracting a CoT from the latent tokens updates the prior probability of correctness
- Passing 3 prompt perturbation tests increases the probability of correctness up to 52 percentage points



¹ Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason E Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. In Second Conference on Language Modeling, 2025.
² Zhenyi Shen, Hanqi Yan, Linhai Zhang, Zhanghao Hu, Yali Du, and Yulan He. 2025. CODI: Compressing Chain-of-Thought into Continuous Space via Self-Distillation. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pages 677–693, Suzhou, China. Association for Computational Linguistics.

² Hanlin Zhu, Shibo Hao, Zhiting Hu, Jiantao Jiao, Stuart Russell, Yuandong Tian. Reasoning by Superposition: A Theoretical Perspective on Chain of Continuous Thought. In The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2025.